ZACHARY A. DAVIS

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Systems & Infrastructure Engineer | CISSP | Active TS/SCI

SUMMARY

Systems engineer with 6 years of experience automating and scaling Windows, cloud, and virtualization infrastructure for federal enterprises (FBI, Joint Staff), including endpoint management across a **150,000-device** fleet and an enterprise vSphere 7-to-8 data center upgrade. Builds production automation in PowerShell, C#, and Python and contributes to full-stack Angular/.NET applications.

TECHNICAL SKILLS

- **Languages:** C#, Python, PowerShell, TypeScript, JavaScript, SQL, C/C++
- **Web & Backend:** Angular, React, .NET / OWIN (Katana), REST APIs, SQL Server 2019/2022, PostgreSQL, MongoDB
- **Cloud & Virtualization:** AWS (S3, EC2), Azure, Hyper-V, VMware vSphere 7/8, Horizon 7/8 (Connection Server)
- **Endpoint & Automation:** MECM/SCCM, HCL BigFix, WSUS, Ansible, DISM, Group Policy, Active Directory, GitLab, Windows Server 2008–2025, RHEL 8
- **Security:** STIG hardening, ACAS/Nessus, PKI/SSL lifecycle, privileged account management, cross-domain solutions

EXPERIENCE

Senior Systems Engineer II (Tier 4, Windows Infrastructure) — **Mar 2024 – Present**
Open Systems Technology, supporting the FBI — Pocatello, ID

- **Led** the enterprise-wide in-place upgrade from Windows 11 23H2 to 25H2 across **~150,000 endpoints** with minimal user disruption, by designing phased MECM deployment rings, automated pre-flight checks, and rollback paths.
- **Automated** the end-to-end Windows image-servicing pipeline for the FBI secure baseline deployed to 150,000 endpoints, cutting build time from **multiple days to one hour**, by scripting offline servicing (DISM, cumulative updates, drivers, hardening) in PowerShell.
- **Engineered** an enterprise troubleshooting feature for a full-stack web application (Angular, C#/.NET, Katana self-hosted, GitLab), reducing mean time to resolution by **45%** for **150,000 users**, by building integrated diagnostic and remediation tooling.
- **Orchestrated** patching and compliance across a 150,000-endpoint fleet with MECM and HCL BigFix, raising patch compliance to **90% within 3 days** of release by automating deployment, reporting, and remediation workflows.
- **Re-architected** the cross-domain transfer process on AWS, eliminating **40+ hours/month** of manual work and disposable-media waste, by replacing physical-media workflows with automated pipelines.
- **Automated** password rotation for 7 privileged service accounts, removing **60 hours/month** of manual effort and reducing errors, by building PowerShell workflows integrated with Active Directory.
- **Built** Hyper-V deployment automation (MECM and SQL Server prerequisites), cutting provisioning from **7 days to 3–4 hours**; migrated 8+ databases from SQL Server 2019 to 2022 and implemented gMSAs to eliminate static credentials.

Systems Engineer II — **Aug 2022 – Feb 2024**
Leidos, supporting the Joint Staff — Hampton Roads, VA

- **Led** the enterprise-wide upgrade from VMware vSphere 7 to vSphere 8 across **48+ Dell PowerEdge ESXi hosts** by planning phased host upgrades, vCenter migration, and hardware/driver compatibility validation.
- **Owned** end-to-end data center operations for the Joint Staff, spanning compute, SAN switch fabric, and Dell PowerStore storage, including storage migrations to PowerStore.
- **Administered** and upgraded (Horizon 7 to 8) two load-balanced VMware Horizon Connection Servers delivering VDI to the **entire Joint Staff** across Hampton Roads and the Pentagon, maintaining high availability through patching and incident response.
- **Automated** SSL certificate lifecycle management in PowerShell, reducing manual renewal effort and preventing expiry-related outages.
- **Led** the disaster-recovery appliance migration from Windows Server 2016 to 2019 with zero data loss, and performed STIG hardening and vulnerability remediation.

System Administrator — **Nov 2020 – Aug 2022**
WR Systems — Little Creek–Fort Story, VA

- **Secured** third-party collaborator assets through ACAS/Nessus scanning and remediation; administered backup and recovery for Windows/Linux users and an Autodesk Vault server, and supported full-site network operations.

PROJECTS

- **Delivery Route Optimizer** (Python, custom hash table, nearest neighbor + 2-opt): Cut fleet mileage 14% (110.2 to 94.7 mi) for a 3-truck, 40-package routing problem using deadline-validated 2-opt optimization and a from-scratch, self-resizing hash table.
- **Health Risk Forecaster** (Python, scikit-learn, pandas): Reduced prediction error 41% vs. a decision-tree baseline (R^2 0.978, 98% recall on high-risk days) by tuning Random Forest and Gradient Boosting ensembles with grid/randomized search and 5-fold cross-validation.
- **Portfolio Website** (React, Vercel, Resend, Cloudflare Turnstile): Designed and deployed a responsive personal site with a serverless contact form and bot verification.

EDUCATION & CERTIFICATIONS

- **B.S., Computer Science**, Western Governors University, 2026 · **CISSP**, 2026 · **CompTIA Security+ CE** · **Active TS/SCI**